

Total Time
78:10
Remaining.

Paper DetailsSection Questions - **15**Section Marks - **15**Total Marks - **50**Total Sections - **3**

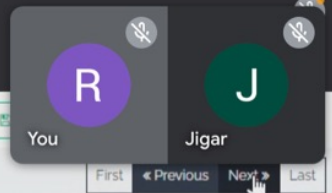
Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design **Language Basics****Language Basics >**1 2 3 4 5 6 **7** 8 9 10 11 12 13 14 15**?** Question**i** Section Instruction

1

Which of the following is not a system call?

☐ chmod☐ getc☐ open☐ lseek



You

Jigar

First < Previous Next > Last

Total Time
78:21
Remaining.

Paper Details

Section Questions - 15

Section Marks - 15

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design Language Basics

Language Basics >

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

 Question Section Instruction

1

The UNIX command `cd ../.`☐ serves no purpose☐ is invalid☐ is same as `cd..`☐ is same as `cd .`

First << Previous Next >> Last

R
You

J
Jigar



Total Time
78:49
Remaining.

Paper DetailsSection Questions - **15**Section Marks - **15**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design **Language Basics**

Language Basics >

1 2 **3** 4 5 6 7 8 9 10 11 12 13 14 15 Question Section Instruction

1

A stack is nothing but a set of

☐ reserved ROM address space☐ reserved RAM address space☐ reserved I / O address space☐ none of the above

First

< Previous

Next >

Last

Total Time
79:04
Remaining.

Paper Details

Section Questions - 15

Section Marks - 15

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design Language Basics

Language Basics >

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

First < Previous Next > Last

Question **Section Instruction**

1

Which of the following statements are correct for the program below?

```
always @(posedge clock) begin
  x = y;
  y = x;
end
```

☐ Swaps the content of x and y☐ Holds value x in both☐ Holds value y in both☐ Holds same value

Total Time
79:20
Remaining.

Paper DetailsSection Questions - **15**Section Marks - **15**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design **Language Basics**

Language Basics >

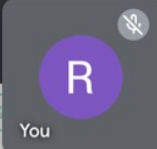
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

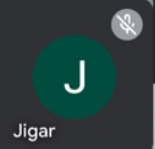
 Question Section Instruction

1

In C language, using goto loop inside for loop is equivalent to using

☐ continue☐ break☐ return☐ None of the above


You


Jigar

First << Previous Next >> Last

Total Time
79:37
Remaining.

Paper DetailsSection Questions - **25**Section Marks - **35**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative **Digital Design** Language Basics**Digital Design >**

6

7

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24

25 Question Section Instruction

1

A traffic signal goes from GREEN to YELLOW, YELLOW to RED and RED to GREEN. In each cycle, GREEN is turned on for 70 seconds, YELLOW is turned on for 5 seconds and RED is turned on for 75 seconds. This traffic light must be implemented using a finite state machine (FSM). The only input to this FSM is a clock of 5 second period. The minimum number of flip-flops required to implement this FSM is

☐

3

☐

4

☐

5

☐

6

Total Time

80:01

Remaining.

Paper Details

Section Questions - 25

Section Marks - 35

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections

+ Quantitative

+ Digital Design

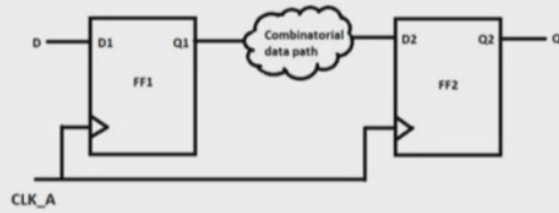
+ Language Basics

Question

Section Instruction

For a given sequential circuit as shown below, assume that both flip flops have a clock to output delay = 10 ns, setup time = 5 ns and hold time = 2 ns. Combinational data path has a delay of 10 ns.

Calculate the maximum frequency of CLK_A for the design to operate correctly

☐ 45.45 MHz☐ 45 MHz☐ 50 MHz☐ 40 MHz

Total Time
80:35
Remaining.

Paper DetailsSection Questions - **25**Section Marks - **35**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative **Digital Design** Language Basics**Digital Design >**

6

7

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9

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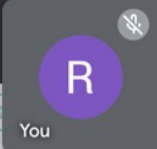
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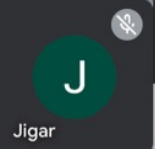
 Question Section Instruction

1

The Fourier series of an odd periodic function contains

☐ odd harmonics only☐ even harmonics only☐ cosine harmonics only☐ sine harmonics only


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First << Previous Next >> Last

Total Time

81:22

Remaining.

Paper Details

Section Questions - 25

Section Marks - 35

Total Marks - 50

Total Sections - 3

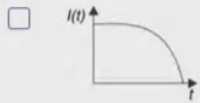
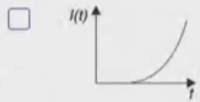
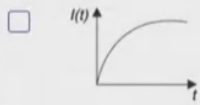
Paper - SNPS_DigiDesign21

Sections

Quantitative

Digital Design

Language Basics

current $i(t)$ is given as $V_s = Ri(t) + 1/c \int_0^t i(t)dt$ Which one of the following represents the current $i(t)$?

Total Time

81:31

Remaining

Paper Details

Section Questions - 25

Section Marks - 35

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections

 Quantitative Digital Design Language Basics

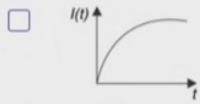
Digital Design > 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

Question Section Instruction

1

A series RC circuit is connected to a DC voltage source at time $t = 0$. The relation between the source voltage V_s , the resistance R , the capacitance C , and the current $I(t)$ is given as

$V_s = RI(t) + 1/C \int_0^t I(t)dt$ Which one of the following represents the current $I(t)$?



Total Time

82:40

Remaining.

Paper Details

Section Questions - 25

Section Marks - 35

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections

 Quantitative Digital Design Language Basics

Digital Design >

2

3

4

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Question

Section Instruction

1

In a series, RLC circuit, $R = 2 \text{ k}\Omega$, $L = 1 \text{ H}$, and $C = 1/400 \text{ }\mu\text{F}$. The resonant frequency is

☐ $2 \times 10^4 \text{ Hz}$ ☐ $1/\pi \times 10^4 \text{ Hz}$ ☐ 10^4 Hz ☐ $2\pi \times 10^4 \text{ Hz}$

First

<< Previous

Next >>

Last

Total Time

81:54

Remaining

Paper Details

Section Questions - 25

Section Marks - 35

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections

 Quantitative Digital Design Language Basics

Digital Design >

5

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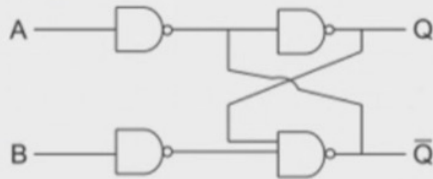
24

Question

Section Instruction

1

The circuit shown in the figure is a

☐ Johnson's counter☐ JK Flip Flop☐ RS Latch☐ Master/Slave D Flip Flop

Total Time

82:13

Remaining

Paper Details

Section Questions - 25

Section Marks - 35

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections

Quantitative

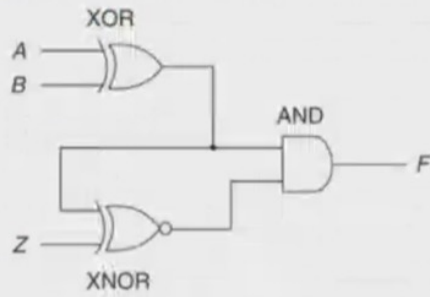
Digital Design

Language Basics

Question

Section Instruction

The output F in the digital logic circuit shown in the figure is



☐ $F = X'YZ + XY'Z$

☐ $F = X'YZ' + XYZ'$

☐ $F = X'Y'Z + XYZ$

☐ $F = X'Y'Z + XYZ'$

Total Time

82:36

Remaining

Paper Details

Section Questions - 25

Section Marks - 35

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections

 Quantitative Digital Design Language Basics

Digital Design >

3

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Question

Section Instruction

1

What does the following code infer in digital design synthesis?

```
always @(posedge clk)
```

```
begin
```

```
  a0 = d;
```

```
  a1 = a0;
```

```
  a2 = a1;
```

```
  q <= a2;
```

```
end
```

☐

A buffer between d and q

☐

A four - stage shift register between d and q

☐

Three buffers between d and a2 followed by a flip - flop between a2 and q

☐

A flip - flop between d and q

R

You

J

Jigar

First

< Previous

Next >

Last



Total Time
83:16
Remaining.

Paper DetailsSection Questions - **25**Section Marks - **35**Total Marks - **50**Total Sections - **3**

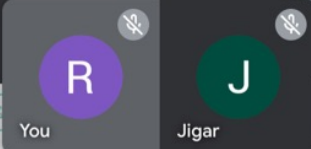
Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design Language Basics**Digital Design >** 1 2 3 4 5 6 7 8 **9** 10 11 12 13 14 15 16 17 18 19 20**?** Question **i** Section Instruction

1

In a BJT circuit a pnp transistor is replaced by npn transistor. To analyze the new circuit

- ☐ all calculations done earlier have to be repeated
- ☐ replace all calculated voltages by reverse values
- ☐ replace all calculated currents by reverse values
- ☐ replace all calculated voltages and currents by reverse values



You

Jigar

First < Previous Next > Last

Total Time

82:58

Remaining

Paper Details

Section Questions - 25

Section Marks - 35

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections

 Quantitative Digital Design Language Basics

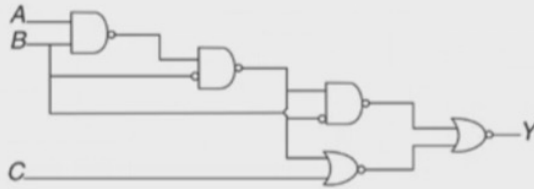
Digital Design > 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

Question

Section Instruction

1

Find Y in the digital logic circuit shown below

☐ $F = A + B + C$ ☐ $F = A$ ☐ $F = B$ ☐ $F = C$

Total Time

83:07

Remaining

Paper Details

Section Questions - 25

Section Marks - 35

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections

 Quantitative Digital Design Language Basics

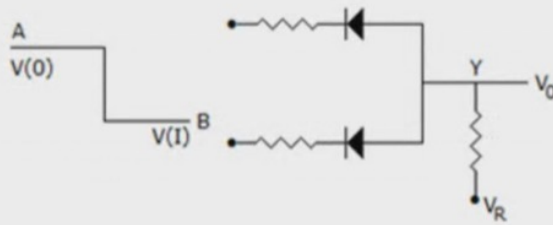
Digital Design > 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

Question

Section Instruction

1

The circuit in the given figure is

☐ positive logic XOR gate☐ negative logic OR gate☐ negative logic AND gate☐ positive logic AND gate

Total Time
83:27
Remaining.

Paper DetailsSection Questions - **25**Section Marks - **35**Total Marks - **50**Total Sections - **3**

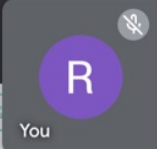
Paper - SNPS_DigiDesign21

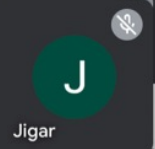
Sections Quantitative **Digital Design** Language Basics**Digital Design >**1 2 3 4 5 6 7 **8** 9 10 11 12 13 14 15 16 17 18 19 20**?** Question**1** Section Instruction

1

The product of the binary numbers 110 and 1010 is

☐ 111100☐ 111110☐ 101011☐ 100010


You


Jigar

First < Previous Next > Last

Total Time
83:38
Remaining.

Paper DetailsSection Questions - **25**Section Marks - **35**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative **Digital Design** Language Basics**Digital Design >**1 2 3 4 5 6 **7** 8 9 10 11 12 13 14 15 16 17 18 19 20**7** Question**1** Section Instruction

1

Negative feedback reduces noise originating at the amplifier input

☐

True

☐

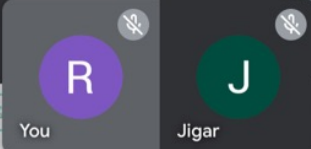
False

☐

Depends on load resistance

☐

Insufficient data



You

Jigar

First < Previous Next > Last

Total Time
83:46
Remaining.

Paper DetailsSection Questions - **25**Section Marks - **35**Total Marks - **50**Total Sections - **3**

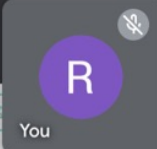
Paper - SNPS_DigiDesign21

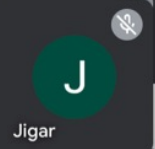
Sections Quantitative Digital Design Language Basics**Digital Design >** 1 2 3 4 5 **6** 7 8 9 10 11 12 13 14 15 16 17 18 19 20**?** Question **i** Section Instruction

1

In Laplace transform, multiplication by e^{-at} in time domain becomes

- ☐ translation by a in s domain
- ☐ translation by $(-a)$ in s domain
- ☐ multiplication by e^{-as} in s domain
- ☐ none of the above


You


Jigar

First < Previous Next > Last

Total Time

83:54

Remaining

Paper Details

Section Questions - 25

Section Marks - 35

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections

 Quantitative Digital Design Language Basics

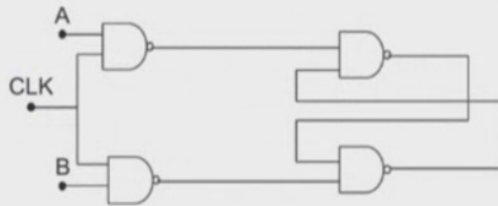
Digital Design > 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

Question

Section Instruction

1

Consider the given circuit. In this circuit, the race around



- ☐ does not occur
- ☐ occurs when CLK = 0
- ☐ occurs when CLK = 1 and A = B = 1
- ☐ occurs when CLK = 1 and A = B = 0

Total Time

84:06

Remaining.

Paper Details

Section Questions - 25

Section Marks - 35

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections

 Quantitative Digital Design Language Basics

Digital Design >

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

Question

Section Instruction

1

A half wave diode circuit using ideal diode has an input voltage $20 \sin \omega t$ volts. Then average and rms values of output voltage are

☐ $10/\pi$ V and 10 V☐ $20/\pi$ V and 10 V☐ $10/\pi$ V and 5 V☐ $20/\pi$ V and 5 V

Total Time

84:13

Remaining

Paper Details

Section Questions - 25

Section Marks - 35

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections

 Quantitative Digital Design Language Basics

Digital Design >

1

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 Question Section Instruction

1

An RC coupled amplifier has an open loop gain of 200 and a lower cutoff frequency of 50 Hz. If negative feedback with $\beta = 0.1$ is used, the lower cut off frequency will be

☐

about 50 Hz

☐

about 5 Hz

☐

about 2.38 Hz

☐

about 70.5 Hz



Total Time

84:27

Remaining

Paper Details

Section Questions - 25

Section Marks - 35

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections

Quantitative

Digital Design

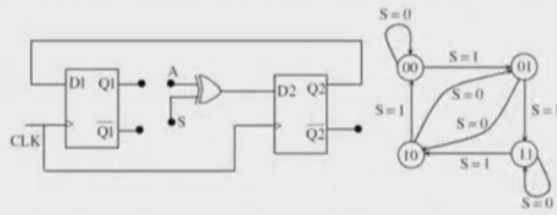
Language Basics

Question

Section Instruction

The digital logic shown in the figure below satisfies the given state diagram when Q1 is connected input A of XOR gate.

Suppose the XOR gate is replaced by a XNOR gate. Which one of the following option preserves the state diagram?



- ☐ Input A is connected to Q'2
- ☐ Input A is connected to Q2
- ☐ Input is connected to Q'1 and S is complemented
- ☐ Input is connected Q'1

Total Time

84:31

Remaining.

Paper Details

Section Questions - 25

Section Marks - 35

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections

Quantitative

Digital Design

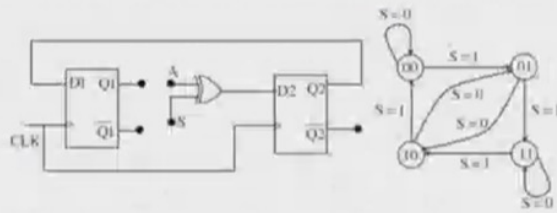
Language Basics

Question

Section Instruction

The digital logic shown in the figure below satisfies the given state diagram when Q1 is connected input A of XOR gate.

Suppose the XOR gate is replaced by a XNOR gate. Which one of the following option preserves the state diagram?



- ☐ Input A is connected to Q'2
- ☐ Input A is connected to Q2
- ☐ Input is connected to Q'1 and S is complemented
- ☐ Input is connected Q'1

Total Time

85:01

Remaining.

Paper Details

Section Questions - 25

Section Marks - 35

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections

 Quantitative Digital Design Language Basics

Digital Design >

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4

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Question

Section Instruction

1

For a system to work, as oscillator the total phase shift of the loop gain must be equal to

☐ 90°☐ 45°☐ 270°☐ 360°

Total Time
89:57
Remaining.

Paper Details

Section Questions - 10

Section Marks - 10

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design Language Basics**Quantitative >** 1 2 3 4 5 6 7 8 9 10**Question** **1** Section Instruction

1

The total number of words that can be made by writing the letters of the word PARAMETER so that no vowel is between two consonants is

☐ 1440☐ 1800☐ 2160☐ 5432

 R
You

 J
Jigar

First < Previous Next > Last

Total Time
89:49
Remaining.

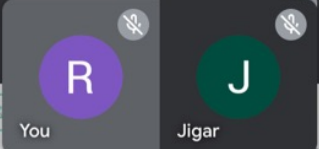
Paper DetailsSection Questions - **10**Section Marks - **10**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design Language Basics**Quantitative >** 1 2 3 4 5 6 7 8 9 10**7** Question **1** Section Instruction

What is probability that the sum of the numbers on three six-sided dice after a throw is "3"

☐ 1/216☐ 1/2☐ 1/36☐ 1/6



You

Jigar

First < Previous Next > Last

1



Total Time
89:42
Remaining.

Paper DetailsSection Questions - **10**Section Marks - **10**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design Language Basics**Quantitative >** 1 2 **3** 4 5 6 7 8 9 10**?** Question **i** Section Instruction

1

In a senary system (base 6), the number 5432 is equal to which of the following numbers in the hexadecimal system?

☐ 1244☐ 4DC☐ 4DD☐ 5432

First << Previous Next >> Last

Total Time
89:36
Remaining.

Paper DetailsSection Questions - **10**Section Marks - **10**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design Language Basics**Quantitative >** 1 2 **3** 4 5 6 7 8 9 10**?** Question **1** Section Instruction

1

In a senary system (base 6), the number 5432 is equal to which of the following numbers in the hexadecimal system?

☐ 1244☐ 4DC☐ 4DD☐ 5432

First < Previous Next > Last

Total Time
89:32
Remaining.

Paper DetailsSection Questions - **10**Section Marks - **10**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections **Quantitative** Digital Design Language Basics**Quantitative >** 1 2 3 **4** 5 6 7 8 9 10**?** Question **1** Section Instruction

Find the odd one out in 8, 27, 64, 100, 125, 216, 343

☐ 25☐ 100☐ 125☐ 343

R
You

J
Jigar

First < Previous Next > Last

1



Total Time
77:03
Remaining.

Paper DetailsSection Questions - **15**Section Marks - **15**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design **Language Basics****Language Basics >**1 2 3 4 5 6 7 8 9 10 11 **12** 13 14 15 Question Section Instruction

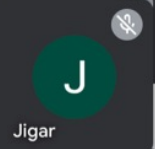
1

Kernel is not involved

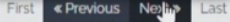
☐ when a read operation is done☐ when a pressed key is echoed on to screen☐ in resource allocation☐ none of the above



You



Jigar





Total Time
76:19
Remaining.

Paper DetailsSection Questions - **15**Section Marks - **15**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design Language Basics

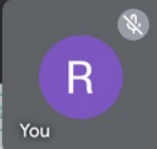
Language Basics >

1 2 3 4 5 6 7 8 9 10 11 12 13 **14** 15 Question Section Instruction

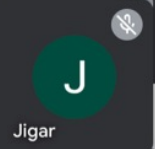
1

In non-blocking assignment, the Verilog compiler

☐ Evaluates all RHS for the current time unit and assign to LHS at the end of time unit☐ Evaluates all RHS for the current time unit and assign to LHS at the current time☐ Both above options are correct, depending on the specific case☐ None of the options are correct



You



Jigar

[First](#) [Previous](#) [Next](#) [Last](#)

Total Time
77:21
Remaining.

Paper DetailsSection Questions - **15**Section Marks - **15**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design **Language Basics****Language Basics >**1 2 3 4 5 6 7 8 9 **10** 11 12 13 14 15**?** Question**i** Section Instruction

1

The following program fragment

`int k = -7;``printf ("%d", 0 < !k);`☐

prints 0

☐

prints a non - zero value

☐

is illegal

☐

prints an illegal value

First

< Previous

Next >

Last

R

You

J

Jigar



Total Time
77:14
Remaining.

Paper Details

Section Questions - 15

Section Marks - 15

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design Language Basics

Language Basics >

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

 Question Section Instruction

1

Select the wrong answer: Shell script is preferable to other forms of programming because it

- ☐ executes faster
- ☐ enhances portability
- ☐ occupies less space
- ☐ makes programming task easier

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Jigar

First < Previous Next > Last

Total Time
76:04
Remaining.

Paper DetailsSection Questions - **15**Section Marks - **15**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design Language Basics

Language Basics >

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

 Question Section Instruction

1

In UNIX, the default permission bits when a file created for the first time is controlled by

☐

chmod value

☐

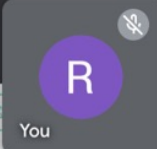
fmask value

☐

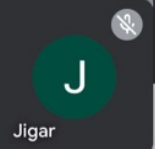
umask value

☐

None of the above



You



Jigar

First < Previous Next > Last

Total Time
76:34
Remaining.

Paper DetailsSection Questions - **15**Section Marks - **15**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design **Language Basics**

Language Basics >

1 2 3 4 5 6 7 8 9 10 11 12 **13** 14 15 Question Section Instruction

1

Which of the following is not a command line delimiter

☐

.

☐

new line

☐

;

☐

&

 R
You

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Jigar

First << Previous Next >> Last

Total Time

78:29

Remaining.

Paper Details

Section Questions - 15

Section Marks - 15

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections

 Quantitative Digital Design Language Basics

Language Basics >

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Question

Section Instruction

1

In Verilog, the default value for reg data type is

☐

0

☐

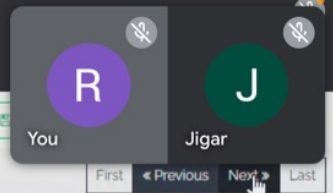
X

☐

Z

☐

1



You

Jigar

First < Previous Next > Last

Total Time
77:54
Remaining.

Paper DetailsSection Questions - **15**Section Marks - **15**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design **Language Basics**

Language Basics >

1 2 3 4 5 6 7 **8** 9 10 11 12 13 14 15 Question Section Instruction

1

If storage class is missing in the array definition, by default it will be taken to be

☐ automatic☐ external☐ static☐ either automatic or external depending on the place of occurrence

First

< Previous

Next >

Last

Total Time
78:41
Remaining.

Paper DetailsSection Questions - **15**Section Marks - **15**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design **Language Basics****Language Basics >**1 2 3 **4** 5 6 7 8 9 10 11 12 13 14 15**?** Question**i** Section Instruction

1

Which of the following statements are true in Verilog?

reg x;

reg y;

assign #10 x = y;

- ☐ Value of Y is assigned to X after delay of 10 units
- ☐ Value of Y is assigned to X and then wait for delay of 10 units
- ☐ Not a valid assignment statement
- ☐ None of the above

R
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Jigar

First < Previous Next > Last

Total Time
80:59
Remaining.

Paper DetailsSection Questions - **25**Section Marks - **35**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design Language Basics**Digital Design >**

6

7

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 Question Section Instruction

1

To prevent a DC return between source and load, it is necessary to use

☐

resistor between source and load

☐

inductor between source and load

☐

capacitor between source and load

☐

either (1) or (2)

Total Time

80:47

Remaining.

Paper Details

Section Questions - 25

Section Marks - 35

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections

 Quantitative Digital Design Language Basics

Digital Design >

6

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 Question Section Instruction

1

A fully charged mobile phone with a 12 V battery is good for a 10-minute talk-time. Assume that, during the talk-time the battery delivers a constant current of 2 A and its voltage drops linearly from 12 V to 10 V as shown in the figure. How much energy does the battery deliver during this talk-time?

☐ 14.4 J☐ 12 kJ☐ 220 J☐ 13.2 kJ

First

<< Previous

Next >>

Last



Total Time

81:14

Remaining

Paper Details

Section Questions - 25

Section Marks - 35

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections

 Quantitative Digital Design Language Basics

Digital Design >

6

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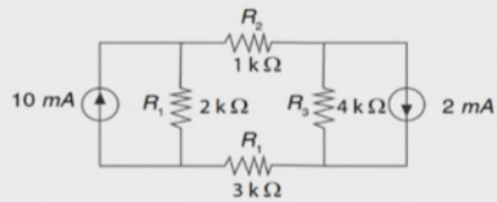
25

Question

Section Instruction

1

The magnitude of current (in mA) through the resistor R_2 in the figure shown is

☐ 2.8 mA☐ 2 mA☐ 10 mA☐ 5 mA

Total Time

89:27

Remaining.

Paper Details

Section Questions - 10

Section Marks - 10

Total Marks - 50

Total Sections - 3

Paper - SNPS_DigiDesign21

Sections

 Quantitative Digital Design Language Basics

Quantitative > 1 2 3 4 5 6 7 8 9 10

Question

Section Instruction

1

If $\log_4 5 = a$ and $\log_5 6 = b$ then $\log_3 2$ is equal to☐ $1/2a+1$ ☐ $1/2b+1$ ☐ $2ab + 1$ ☐ $1/2ab-1$

First < Previous Next > Last

Trying to speak? Turn on your microphone.



 R
You

 J
Jigar

Total Time
89:19
Remaining.

Paper DetailsSection Questions - **10**Section Marks - **10**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design Language Basics**Quantitative >** 1 2 3 4 5 **6** 7 8 9 10 Question Section Instruction

1

From a point P on a level ground, the angle of elevation of the top tower is 30° . If the tower is 100 m high, the distance of point P from the foot of the tower in meters is

☐ 149☐ 156☐ 173☐ 200

R
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First < Previous Next > Last

Total Time
89:11
Remaining.

Paper DetailsSection Questions - **10**Section Marks - **10**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design Language Basics**Quantitative >** 1 2 3 4 5 6 **7** 8 9 10**?** Question **1** Section Instruction

1

Let $t_n = n(n!)$ then $\sum_{n=1}^{15} t_n$ is equal to☐ $15! - 1$ ☐ $15! + 1$ ☐ $16! - 1$ ☐ $16!$

First < Previous Next > Last

Total Time
89:05
Remaining.

Paper DetailsSection Questions - **10**Section Marks - **10**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design Language Basics**Quantitative >** 1 2 3 4 5 6 7 **8** 9 10**?** Question **i** Section Instruction

1

A can finish a work in 18 days and B can do the same work in 15 days. B worked for 10 days and left the job. In how many days, A alone can finish the remaining work?

☐ 5☐ $5\frac{1}{2}$ ☐ 6☐ 8

First < Previous Next > Last

Total Time
88:57
Remaining.

Paper DetailsSection Questions - **10**Section Marks - **10**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design Language Basics**Quantitative >** 1 2 3 4 5 6 7 8 **9** 10**?** Question **1** Section Instruction

1

If a person walks at 14 km/hr instead of 10 km/hr, he would have walked 20 km more. The actual distance travelled by him is

☐ 50 km☐ 56 km☐ 70 km☐ 80 km

First < Previous Next > Last

Total Time
88:48
Remaining.

Paper DetailsSection Questions - **10**Section Marks - **10**Total Marks - **50**Total Sections - **3**

Paper - SNPS_DigiDesign21

Sections Quantitative Digital Design Language Basics**Quantitative >** 1 2 3 4 5 6 7 8 9 **10****?** Question **1** Section Instruction

1

Six bells commence tolling together and toll at intervals of 2, 4, 6, 8, 10 and 12 seconds respectively. In 30 minutes, how many times do they toll together ?

☐ 4☐ 10☐ 15☐ 16

First < Previous Next > Last